

Figure 14: **Pathway edit distance and pretraining objective on nanoMoE during pretraining.** The left panel reports the overall pathway edit-distance trajectory across checkpoints without grouping samples by their memorization steps. The right panel instead visualizes pathway dynamics for two memorized data groups (memorized at 30k and 45k steps). After their respective memorization points, both groups exhibit a consistent post-memorization decline in edit distance despite a plateaued pretraining objective, indicating that nanoMoE continues to reorganize routing pathways toward more structured and shared computation. This mirrors the memorization-to-generalization transition observed in large-scale OLMoE.

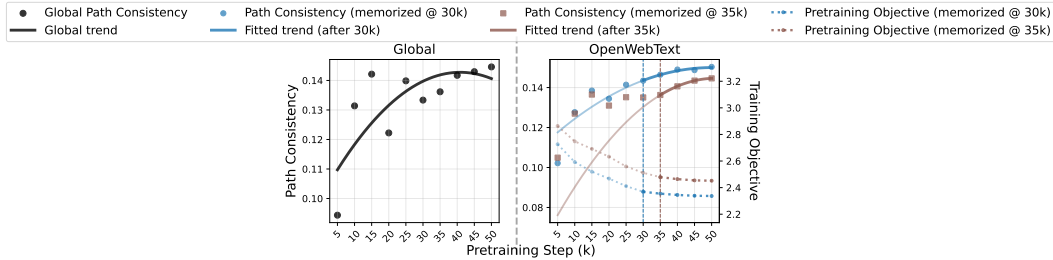


Figure 15: **Evolution of pathway consistency in nanoMoE.** The left panel reports the overall progression of pathway consistency across checkpoints. The right panel further analyzes two memorized data groups (memorized at 30k and 35k steps), showing that once memorization is achieved, each group undergoes continued refinement of its layer-to-layer routing transitions.

LLM Usage Statement

In preparing this manuscript, we used LLM solely as a writing assistance tool. Specifically, the model was employed to aid in grammar correction, sentence refinement, and improving clarity of exposition. It did not contribute to research ideation, experimental design, data analysis, or interpretation of results.

Table 5: **Correlation Between Pathway Metrics and Downstream Accuracy on nanoMoE** across four commonsense benchmarks (ARC-Easy, ARC-Challenge, HellaSwag, OpenBookQA). p -value:

$p < 0.1$, $p < 0.05$, $p < 0.01$, $p > 0.1$, $p > 0.5$, $p > 0.8$. Preferred correlation: positive , negative .

Metric	ARC-Easy		ARC-Challenge		HellaSwag		OpenBookQA	
	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman
Pathway Distance (pairwise)	-0.8113	-1.0000	-1.0000	-1.0000	-0.9713	-1.0000	-0.8366	-0.9000
Pathway Consistency (sample-wise)	0.9416	1.0000	0.8818	1.0000	0.9234	0.8000	0.9979	1.0000
Training Loss	0.3301	-0.1000	-0.7799	-0.5000	-0.2182	-0.2000	0.5984	0.8000
Training Loss (moving avg.)	0.4664	0.4000	-0.6078	-0.8000	-0.3580	0.2000	-0.8655	-0.4000